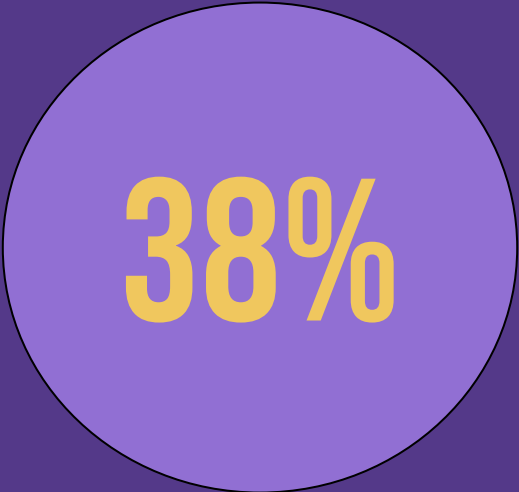




38%

The percent increase in Domestic Abuse rates
when England lose a match¹

1. Full Fact. "World Cup: Does Domestic Abuse Spike When England Lose?" Full Fact, 19 June 2018, fullfact.org/crime/world-cup-domestic-abuse/.

Group 1

BEYOND THE COURT: EXPLORING PATTERNS BETWEEN LAKERS PERFORMANCE AND DOMESTIC VIOLENCE INCIDENTS

*Jacob Dyer, Riley Conner, Leela Maitra, Ryo Suguwara, Sydney
Schaefer, Tess Palmer*

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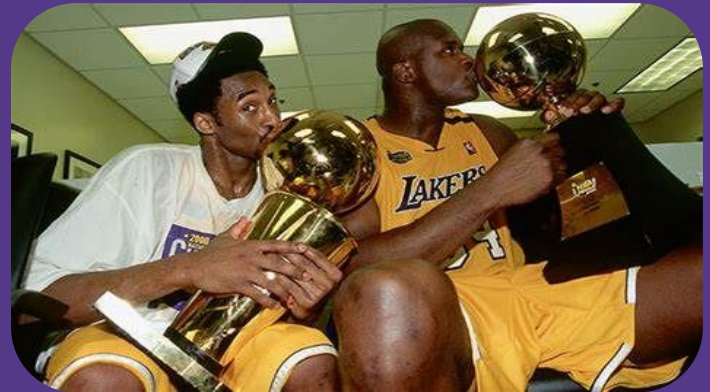
6. Conclusion

INTRODUCTION

The Lakers: A Culturally Iconic Franchise Intertwined with Los Angeles Identity

Subsequent **Game Outcomes** Can Elicit **Intense Emotional Responses**

Research Indicates a **Rise in Domestic Violence Reports** During Major Sporting Events



RESEARCH QUESTION

How does a Los Angeles Lakers game—and the game's outcome—affect the daily volume of domestic-violence calls in Los Angeles?



DATA COLLECTION + PREPARATION

- **Lakers game stats**

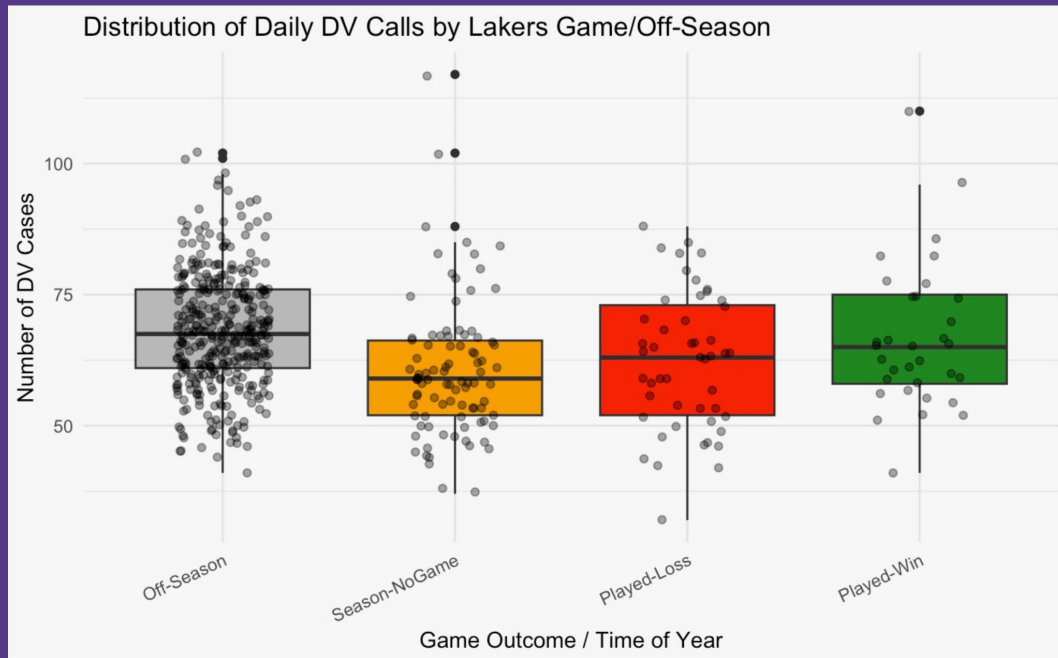
- 2021-2022 regular season game outcomes and streaks (from Basketball Reference)

- **Crime**

- Daily LAPD incident records (2020–present), filtered for six domestic violence-related crime codes using the LA Open Data Portal's UCR guidelines

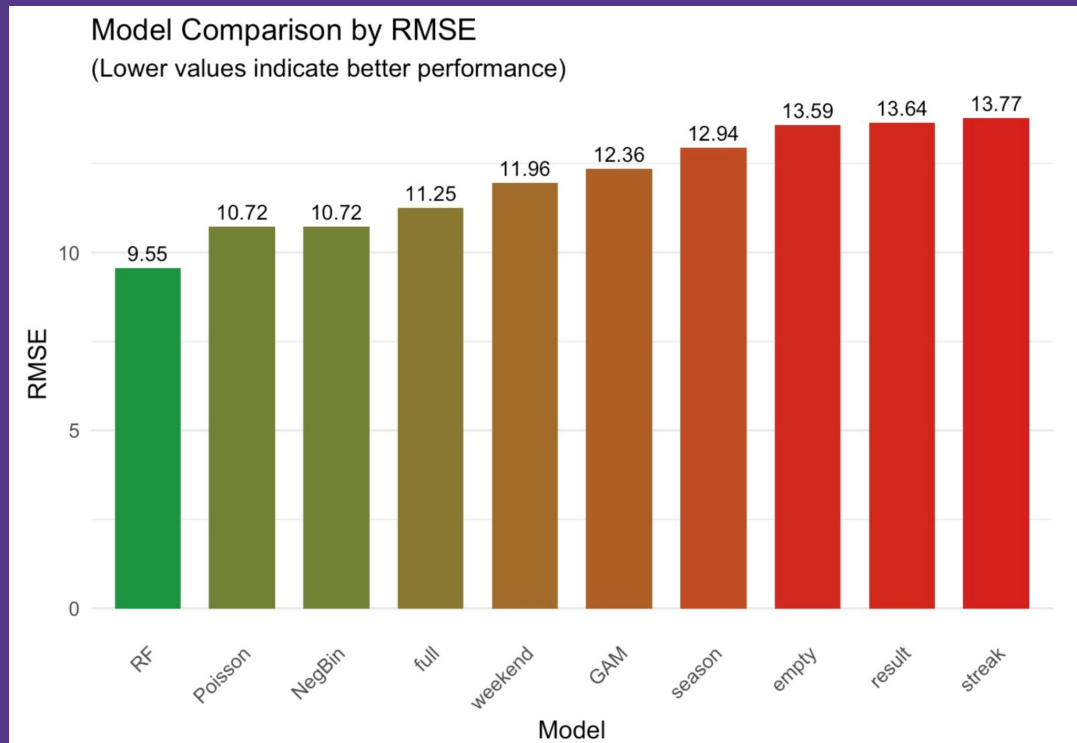
- **Key variables**

- *DV Count and Period2*



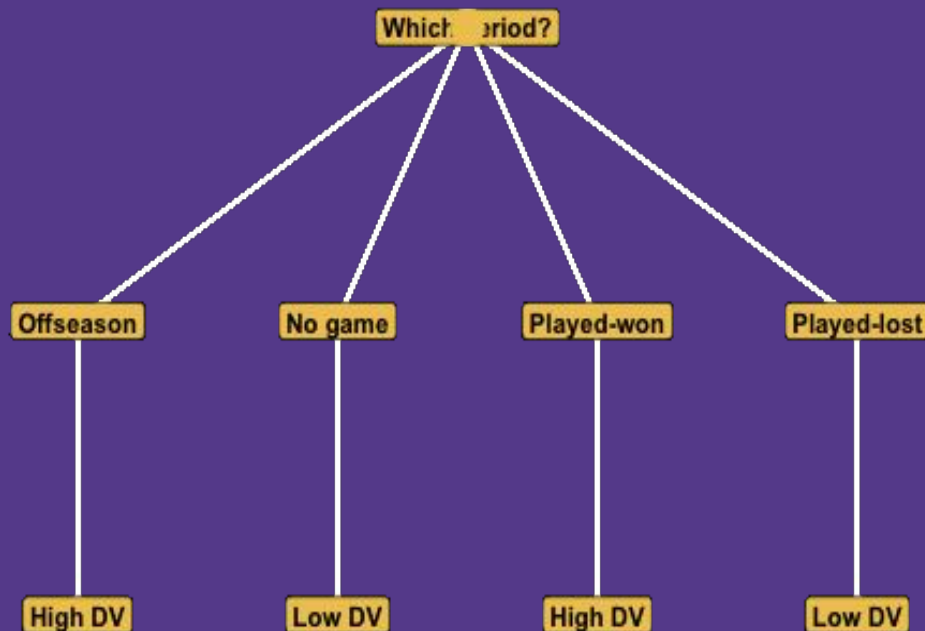
MODELING

- **Tested 10 models** to predict domestic violence counts from Lakers game data
- Used **Poisson regression** because our response variable is a count
- Trained and tested with **10-fold cross validation**



THE WINNING MODEL

- Each train point is asked a series of questions
- Data is grouped into “leaf nodes” based on their answers
- Leaf nodes are labelled as ‘High DV’ or ‘Low DV’
- Test data follows the same path and is classified by the leaf it lands in



CLASSIFICATION SUMMARY

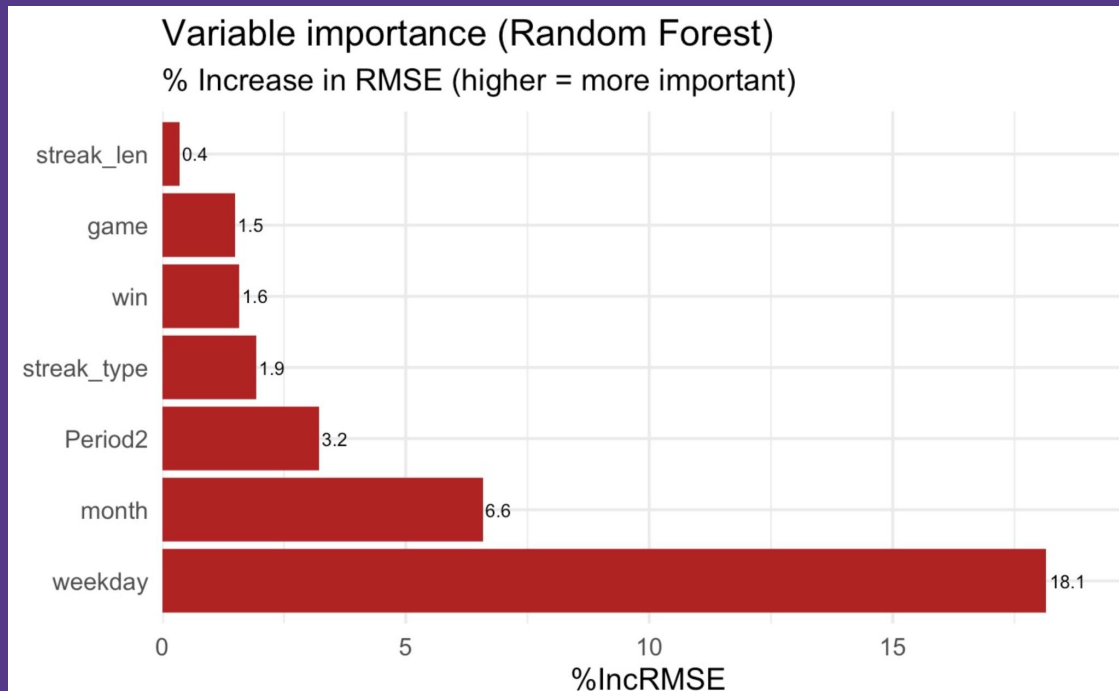
- **Specificity is 0.93**, so the model avoids false positives well
- **Sensitivity is 0.53**, so the model only correctly classifies a *High DV day half of the time*

Confusion Matrix

Predicted DV Day Type	Actual DV Day Type	
	High DV	Low DV
Low DV	4	51
High DV	28	25

KEY FINDINGS

- **Calendar variables** like *weekday* and *month* are the strongest predictors
- Lakers related variables like *outcome* and *streak* have **little to no predictive power**



CONCLUSION

Limitations:

Missing alcohol as a key confounding variable

Future Studies:

Find alcohol data and focus on more crime types or more sports

Takeaways:

Challenging assumptions that sports events lead to DV spikes

Implications:

LAPD should allocate resources based on calendar schedules



THANK YOU!



SOURCES

- Card, D., & Dahl, G. B. (2011). Family violence and football: The effect of unexpected emotional cues on violent behavior (No. w15497). National Bureau of Economic Research. **<https://davidcard.berkeley.edu/papers/card-dahl-family-violence.pdf>**
- Cardazzi, A., McCannon, B. C., Humphreys, B. R., & Rodriguez, Z. (2022, July 29). Emotional cues and violent behavior: Unexpected basketball losses increase incidents of family violence. *Journal of Law, Economics, and Organization*, 40(2), 267–288. **<https://doi.org/10.1093/jleo/ewad017>**
- Castleman, T. (2023, October 6). A troubling California trend: More violent crimes with guns even as restrictions tighten. *Los Angeles Times*. **<https://www.latimes.com/california/story/2023-10-06/a-troubling-trend-in-california-more-violent-crimes-are-being-committed-with-guns-even-as-restrictions-tighten>**
- Federal Bureau of Investigation. (2018, June 26). Uniform Crime Reporting (UCR) Program. U.S. Department of Justice. **<https://www.fbi.gov/services/cjis/ucr>**